way we can decide the size of the Bloom filter's buffer and the number of hash functions needed to have optimal precision.

There are three variables in the formula for `p(n, m, k)`:

- `m`, the number of bits in the buffer.
- `n`, the number of elements that will be stored in the container.
- `k`, the number of hash functions.

Of those three, `k` is the one that seems less meaningful to us. Or, from another point of view, it is the one less coupled with our problem. In fact, `n` is probably going to be a variable that can we can estimate, but we can’t fully control: we might need to store as many elements as we get requests for, but most of the times we can anticipate the volume of requests, and make estimates pessimistic, to be on the safe side.

We might be limited in the choice of `m` as well, since we could have memory constraints: maybe we can't use more than `m` bits.

For `k`, instead, we have no constraint on it, and we can tune it to get optimal precision meaning, as we explained in appendix F, minimal probability of false positives.

Luckily finding the optimal value for `k`, given `m` and `n`, isn't even that hard: it's just a matter of finding the minimum for the function